Naked-Eye Observations: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — Hand angles. Return →

Six fist widths span $6 \times 10^\circ = 60^\circ$. **Check:** the hand scale is approximate; individual calibration is needed for precise work.

Solution 2 — Magnitude difference. Return →

The difference is 5 mag, so the flux ratio is $10^{0.4 \times 5} = 100$. Star A is 100 times brighter. **Check:** smaller magnitude means brighter.

Solution 3 — Star count. Return →

The ratio is $10^{0.6(6-3)} = 10^{1.8} = 63$. **Check:** a three-magnitude improvement reveals far more than twice as many stars because the magnitude scale is logarithmic.

Solution 4 — Dark adaptation. Return →

White light bleaches rod photopigments and undoes much of the dark adaptation, which takes tens of minutes to rebuild. Averted vision looks slightly away from a faint object so its light falls on the rod-rich

peripheral retina, which is more sensitive in dim light. **Check:** faint objects often disappear when stared at directly.

Solution 5 — Angular separation. Return →

$28^\circ / 5^\circ = 5.6$ pointer separations. **Check:** this is the familiar "follow the pointers about five and a half times" rule for finding Polaris.

Solution 6 — Meteor rate. Return →

The rate is $12/(40/60) = 18$ meteors/hour. This is not a true zenithal hourly rate because the radiant may not be at the zenith, the limiting magnitude may differ from the standard, and the observer's field of view may be obstructed. **Check:** raw counts must be corrected for geometry and sky conditions.

Solution 7 — Log entries. Return →

A good log includes date and time, location, sky conditions and limiting magnitude, target and method, comparison stars or angular references, and a sketch or description with uncertainty. **Check:** another observer should be able to understand and repeat the observation.

Solution 8 — Variable-star estimate. Return →

If "halfway" means the arithmetic mean of the two comparison fluxes, the magnitude is $-2.5 \log_{10}[(10^{-0.4\times5.0} + 10^{-0.4\times5.6})/2] = \mathbf{5.26}$ mag. A midpoint in magnitude would instead be 5.30 mag. **Check:** visual estimates are approximate; record the comparison stars and state what "halfway" means.

Solution 9 — Moon size. Return →

$10^\circ / 0.5^\circ = 20$ Moon diameters. **Check:** the Moon is much smaller than most beginners expect; a fist covers many lunar disks.

Solution 10 — Light-pollution estimate. Return →

The factor is $10^{0.6(6.0-3.5)} = 10^{1.5} = 31.6$. The model assumes a uniform Euclidean distribution of stars and no extinction or Galactic structure. **Check:** the real increase is large but not exactly this factor because the sky is not uniform.